

American Pest

MOA / IRAC Quick Reference Guide

Product Rotation Cheat Sheet for Technicians

Why This Matters

Repeated use of the same Mode of Action (MOA) can lead to:

- Product resistance
- Poor control
- Repeat callbacks
- Reduced bait acceptance
- Increased customer complaints

The goal is NOT just to kill pests today. The goal is long-term control.

Quick Rules

DO:

- Rotate between DIFFERENT IRAC groups
- Use IGRs to support rotations
- Document what chemistry was used
- Think long-term on recurring accounts

DO NOT:

- Keep switching brands within the SAME MOA group
 - Repeatedly use pyrethroids on recurring issues
 - Ignore resistance signs
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Common IRAC / MOA Groups We Use

IRAC Group	Chemistry Type	Common Products	Notes
3A	Pyrethroids	Demand CS, CyKick CS, Shockwave 1, Bifen I/T	Fast knockdown. Avoid repeated use.
4A	Neonicotinoids	Alpine WSG, PT Alpine Pressurized	Excellent non-repellent rotation option
22A	Indoxacarb	Advion Roach Gel, Advion Ant Gel	Great for resistant roach and ant populations
13	Chlorfenapyr	Phantom	Excellent non-repellent rotation chemistry
1B	Organophosphates	Niban-FG (stomach poison classification support)	Different chemistry support option
6	Avermectins	Advion WDG	Excellent non-repellent chemistry
7C	Juvenile Hormone Mimic (IGR)	NyGuard IGR	Supports long-term control
15	Chitin Synthesis Inhibitor	Gentrol IGR	Prevents immature development
16	Chitin Synthesis Inhibitor	Tekko Pro	Dual-action IGR
UN	Borates / Other	Nibor-D, Boractin	Different chemistry option

Residential & Commercial Product Quick Reference

Group 3A — Pyrethroids

Use Carefully

Product	Primary Use
Demand CS	General pest / perimeter
CyKick CS	Exterior and crack & crevice

Product	Primary Use
Shockwave 1	Flush-out / aerosol cleanout
Bifen I/T	Exterior perimeter
Transport Mikron	General pest and mosquito

Rotation Tip

Do NOT repeatedly rotate between pyrethroids only. Example:

- Demand CS → CyKick CS → Bifen I/T

This is still Group 3A chemistry.

Group 4A — Neonicotinoids

Excellent Rotation Chemistry

Product	Primary Use
Alpine WSG	Roaches, ants, commercial accounts
PT Alpine Pressurized	Crack & crevice aerosol
Alpine Fly Bait	Fly control

Rotation Tip

Excellent follow-up after repeated pyrethroid use.

Residential Service Rotation

Standard Residential Exterior / Interior Rotation

Rotation Option 1

- Alpine WSG (4A)
- Advion Ant Gel (22A)
- NyGuard IGR (7C)

Rotation Option 2

- Demand CS (3A)

- Ant dust in wall voids
- Tekko Pro (16 + 7C)

Rotation Option 3

- Phantom (13)
- Advion Ant Gel (22A)
- Light dust applications in voids/cracks

Goal

Avoid repeatedly spraying only pyrethroids every service.

Ant Gel & Bait Rotation

Product	IRAC Group	Best Use
Advion Ant Gel	22A	General ant control
Maxforce Quantum	20A	Sweet-feeding ants
Optigard Ant Gel	3A metabolic disruptor profile	Sensitive interior accounts
Advance 375A	Metabolic bait	Protein-feeding ants

Rotation Tip

If ants stop feeding on one bait:

- switch food type
- OR switch MOA group
- OR both

Dust & Void Treatment Options

Product	Chemistry / MOA	Best Use
DeltaDust	3A	Wall voids / electrical
Drione Dust	Silica + Pyrethrin blend	Fast dry void work
Boractin	Borate	Wood voids / carpenter ants
Nibor-D	Borate	Moisture areas / drains

Product	Chemistry / MOA	Best Use
Cimexa Dust	Amorphous silica gel / mechanical desiccant	Bed bugs, voids, long-term dry applications

Cimexa Notes

- Excellent for bed bug work
- Long residual in dry voids
- Non-repellent mechanical kill
- Works differently than traditional insecticides

Important

Dusts should be:

- light
- targeted
- hidden

Heavy visible dusting = poor application.

Group 22A — Indoxacarb

Resistance Management Gold

Product	Primary Use
Advion Roach Gel	German roaches
Advion Ant Gel	Ant control

Rotation Tip

Excellent for resistant accounts. Works very well in rotation with Alpine products.

IGR Products

These Support EVERYTHING

Product	IRAC Group	Purpose
NyGuard IGR	7C	Breaks reproductive cycle

Product	IRAC Group	Purpose
Gentrol	15	Stops immature development
Tekko Pro	16 + 7C	Multi-stage growth control

Important

IGRs are NOT stand-alone cleanout products. They SUPPORT long-term control.

Example Rotation Strategy

Restaurant German Roach Account

Initial Service

- Alpine WSG (4A)
- Advion Roach Gel (22A)
- Gentrol (15)

Follow-Up Service

- PT Alpine Pressurized (4A)
- Different bait placements
- Continue IGR

Heavy Cleanout Only If Needed

- Shockwave 1 (3A)

Avoid

Repeated monthly pyrethroid treatments.

Resistance Warning Signs

Watch for:

- Roaches surviving direct contact
 - Reduced bait feeding
 - Faster population rebound
 - Recurring activity after treatment
 - Heavy callbacks on previously treated accounts
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Smart Technician Mindset

The best technician is NOT:

- the one using the strongest product

The best technician IS:

- the one using the RIGHT chemistry at the RIGHT time
 - while rotating intelligently
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Final Takeaways

Remember:

- Rotate MOA groups
 - Use IGRs regularly
 - Avoid repeated pyrethroid-only programs
 - Think long-term
 - Document treatments properly
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